# Ziply

*AI-Powered Website Generator for Nonprofits*

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# The Problem

**~1.9M+**

**Nonprofits in the U.S.***

*Each needs a web presence*

***Candid Learning*

**63%**

**Prefer to Donate Online***

*A bad website costs you donors*

***Double the Donation*

**42%**

**Lack Adequate Resources***

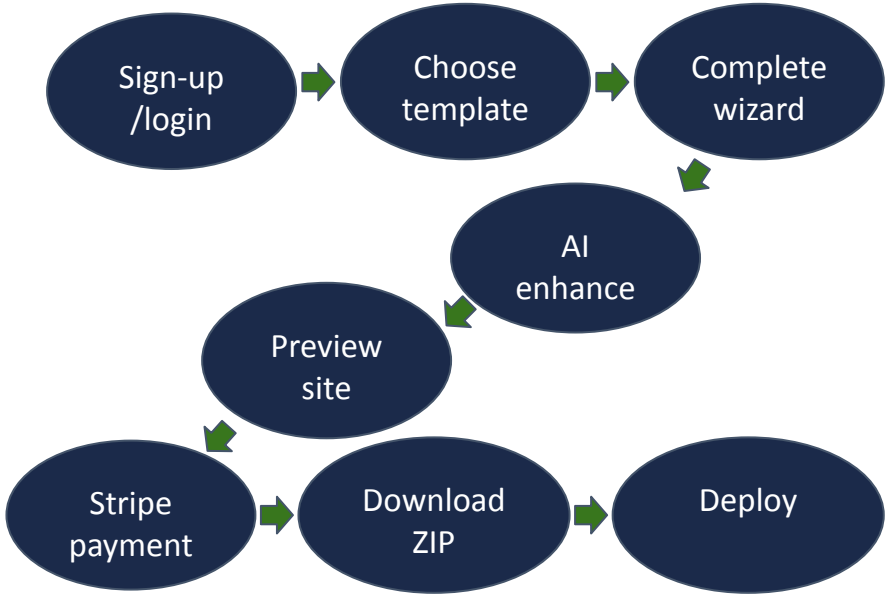
*Can't afford tech staff*

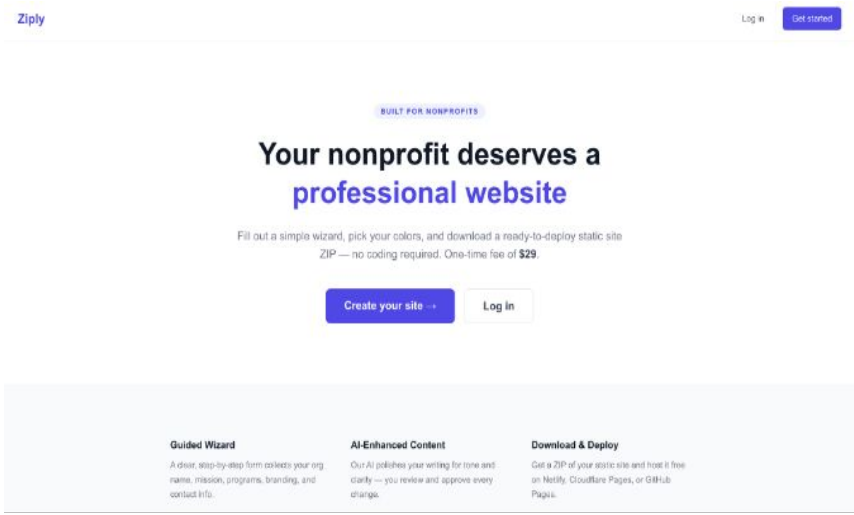
***Forvis Mazars*

*Small nonprofits need a professional web presence — but can't afford agencies, and don't have technical staff.*

# What is Ziply?

A full-stack web app that guides nonprofits through a multi-step wizard to generate a complete, deployable static website with no technical expertise required.



# Tech Stack

*Built on cloud-native, production-grade tools — the same stack used in enterprise software.*

## Frontend

*What the user sees and interacts with*

- Next.js 14
- TypeScript
- Tailwind CSS

## Backend & Data

*The engine running behind the scenes*

- Node.js + Express
- Azure Cosmos DB
- Azure Blob Storage
- Handlebars Templates

## External Services

*Third-party integrations we plugged in*

- Stripe Checkout + Webhooks
- OpenAI GPT API

**Every layer chosen for reliability, scalability, and real-world production use.**

# Our Journey

## Planning & Setup

- Defined scope & SaaS architecture
- Read over documentation on any unfamiliarities within tech stack
- Project skeleton + GitHub + Azure setup
- Handlebars vs. EJS prototyping
- CI/CD pipeline configured

## Building the MVP

- JWT auth (register/login/logout)
- Template engine V1 + wizard UI
- AI content enhancement (GPT)
- ZIP generation + Azure Blob/Cosmos DB
- Stripe Checkout + webhook integration

## Expanding the Product

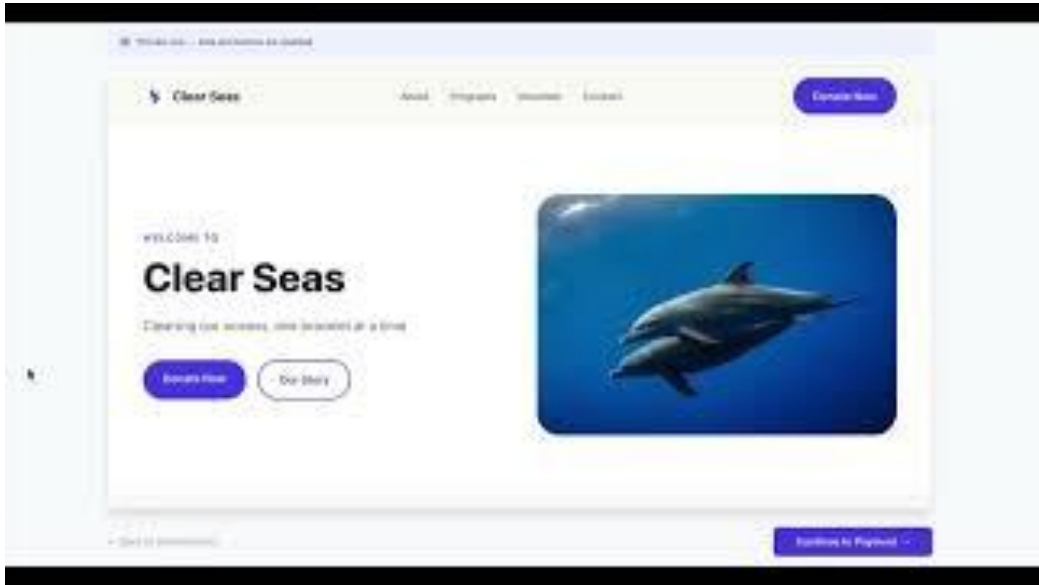
- 3 templates added (basic, modern, warm)
- Image upload to Azure Blob Storage
- FlowNav + Spinner UI components
- 403 blob access bug resolved

## Polish & Ship

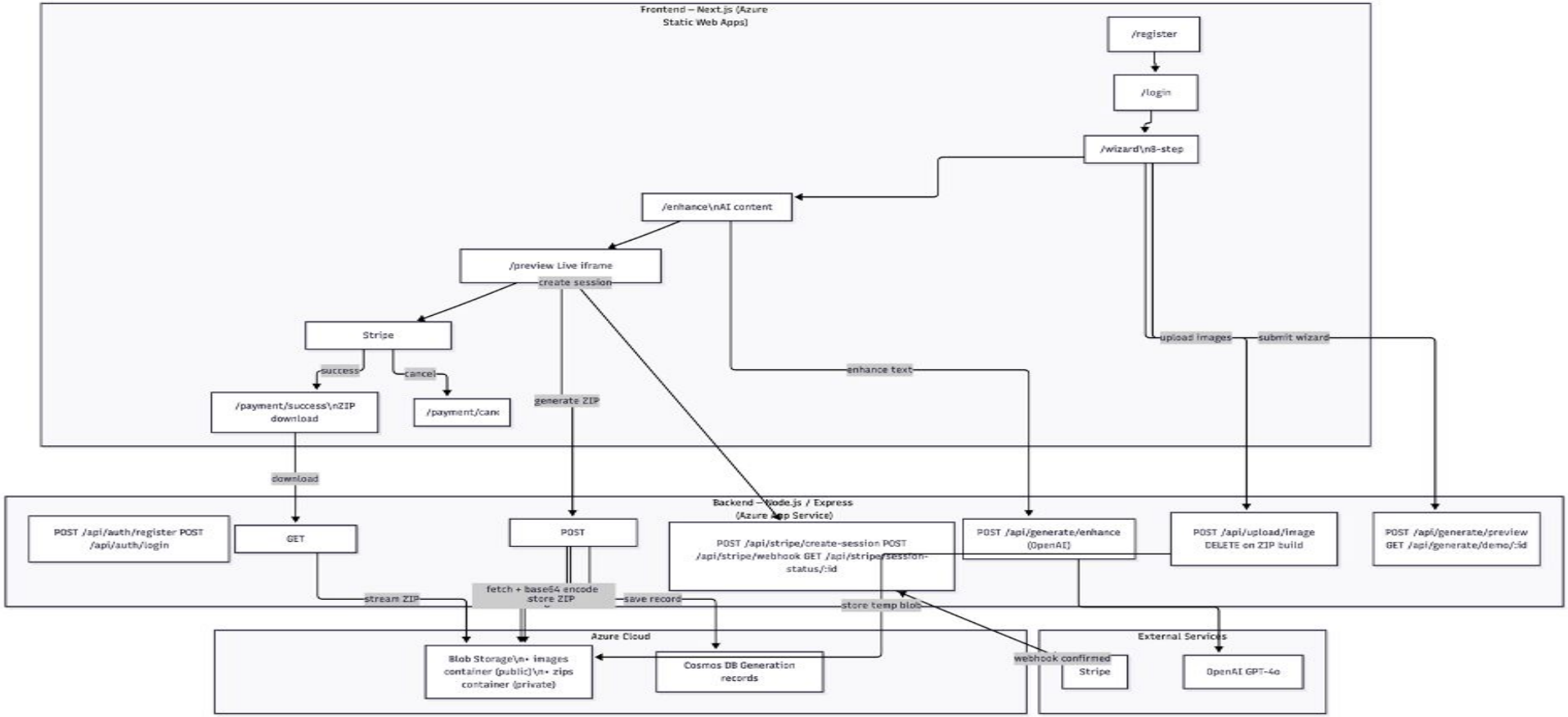
- Base64 image embedding
- Hero + programs image slots (all 3 templates)
- Template preview model
- Netlify + GitHub Pages deploy guides

# Video Demo

https://www.youtube.com/watch?v=ESS25QmXyBI



# System Architecture



# What We Built vs. Original Scope

## Delivered

- 3 visually distinct templates
- 8-step guided wizard
- AI content enhancement (GPT)
- Multi-slot image upload (hero, about, programs)
- Logo upload across all templates
- Base64 self-contained ZIP
- Stripe one-time payment ($29)
- Template preview modal
- Netlify + GitHub Pages deploy guides

## Scope Adjustments

- 3 templates (planned 4–6)
- **Depth > breadth trade-off**
- No user dashboard / re-download
- Stripe still in test mode
- Must complete the entire wizard each time you edit your website

*Decision: investing sprint time in **polishing existing features** delivered more value than adding a 4th template.*

# What We Learned

## Hard Skills

- **Azure Blob Storage (upload, stream, delete)**
- **Azure Cosmos DB**
- **Azure Static Web App Deployment**
- Base64 image encoding & data URIs
- Multi-step form state management in React
- OpenAI API prompt engineering
- Handlebars server-side templating
- Stripe Checkout + webhook verification

## Soft Skills & Lessons

- Scope discipline: finishing > starting
- Working feature ≠ production-ready feature
- Branch-level ownership prevents conflicts
- Communication cadence matters on short sprints
- **The user's journey after download is part of the product**

# If We Had More Time…

### 📂 User Dashboard

Purchase history, re-download past ZIPs, manage organization info

### 🎨 More Industries

Healthcare, education, faith-based orgs — specialized layouts for each sector

### ⚡ Live Customization

Change colors or fonts in-browser without rerunning the full wizard

### 🚀 One-Click Deploy

Direct Netlify integration — click a button and your site goes live from the app

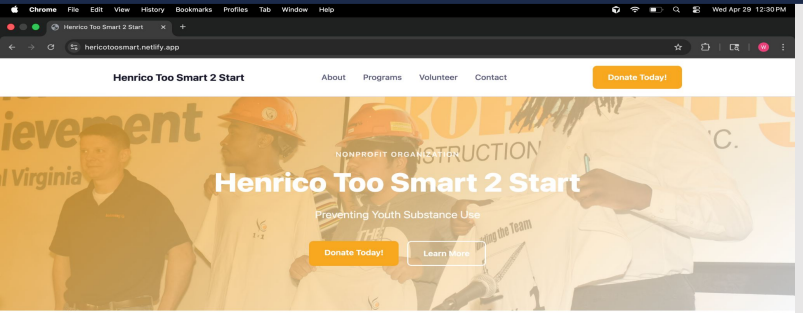
### 📄 Multi-Page Sites

Separate About, Donate, and Events pages for larger nonprofits

### 💳 Production Stripe

Remove test mode — real $29 charge and launch commercially

# Octavia Marsh - Henrico Too Smart 2 Start




OUR MISSION

## Why We Exist

Our mission is to reduce substance use among youth in Henrico County.

Henrico Too Smart 2 Start | About | Programs | Volunteer | Contact | Donate Today!

## Our Programs

Explore the ways we serve our community.

### Drive Smart Henrico

A program of the Henrico Too Smart 2 Start Coalition that works to prevent impaired driving among individuals aged 16 to 21 by promoting sober decision-making and safe transportation options. Through education, community pledges, and awareness campaigns, the program encourages

### Youth Ambassadors Program

Engages high school students in grades 9 through 12 in leadership and substance abuse prevention, empowering youth to plan and lead community initiatives.

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ABOUT US

## Who We Are

The Henrico Too Smart 2 Start Coalition is a nonprofit organization dedicated to promoting health and wellness in Henrico County by reducing youth substance abuse through education, advocacy, and community collaboration. Since becoming an independent entity in 2014, the Coalition has focused on preventing the early use of substances such as alcohol, nicotine, and marijuana while leading county-wide awareness and prevention initiatives.

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## Support Our Work

Every contribution helps us further our mission and serve more people in need. Your generosity makes a real difference.

Donate Today!

## Volunteer With Us

Contact us today to learn how you can volunteer!

Sign Up to Volunteer

# Thank You

*Questions?*